

# MAT 426: Topics in Linear Algebra

## Exam 2

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| Points: | 16 | 19 | 12 | 11 | 12 | 18 | 12 | 100   |
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- *You are NOT ALLOWED to use your book, notes and homework*
- *You are NOT ALLOWED use internet/web resources.*
- *You are NOT ALLOWED to consult with others.*
- *You are allowed to use a scientific, non graphing calculator.*

### Honesty Pledge

I affirm that the work I am submitting upholds the Centre College standard for academic honesty, that it is solely my own work, and that I have neither given nor received any unauthorized assistance from other people or resources.

Solutions

Printed Name

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1. (6 points) Answer the following by circling either TRUE or FALSE.

True ☒ False Every matrix has an orthogonal diagonalization.

☒ True False Gram-Schmidt orthogonalization takes a basis for a subspace and produces an orthogonal basis for the same subspace.

True ☒ False Gram-Schmidt and QR via Householder matrices always produce an orthogonal matrix  $Q$  of the same size.

☒ True False When implemented in a computer, the vector  $\mathbf{x}_i$  used to create zeros in column  $i$  using Householder matrices is typically normalized by dividing by the largest entry in  $\mathbf{x}_i$ .

☒ True False The eigenvalues of a Householder matrix  $H$  are  $\pm 1$ .

True ☒ False The singular values of a matrix  $A$  can be negative.

2. (10 points) Let  $W = \text{Span} \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}$ . Write  $\mathbf{y} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 4 \end{bmatrix}$  as the sum of a vector in  $W$  and a vector orthogonal to  $W$ .

$$\begin{aligned} \hat{\mathbf{y}} &= \left( \frac{\bar{\mathbf{y}} \cdot \bar{\mathbf{u}}_1}{\bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_1} \right) \bar{\mathbf{u}}_1 + \left( \frac{\bar{\mathbf{y}} \cdot \bar{\mathbf{u}}_2}{\bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_2} \right) \bar{\mathbf{u}}_2 + \left( \frac{\bar{\mathbf{y}} \cdot \bar{\mathbf{u}}_3}{\bar{\mathbf{u}}_3 \cdot \bar{\mathbf{u}}_3} \right) \bar{\mathbf{u}}_3 \\ &= \frac{3}{3} \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} + \frac{12}{3} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} + \frac{-3}{3} \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \\ 4 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ 2 \\ 3 \\ 4 \end{bmatrix} \end{aligned}$$

$$\bar{\mathbf{z}} = \bar{\mathbf{y}} - \hat{\mathbf{y}} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 4 \end{bmatrix} - \begin{bmatrix} 5 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

$$\bar{\mathbf{y}} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 4 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

3. (10 points) Find the matrix  $Q$  in the  $QR$  decomposition of the matrix  $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ .

$$\bar{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\bar{v}_2 = \bar{x}_2 - \left( \frac{\bar{x}_2 \cdot \bar{v}_1}{\bar{v}_1 \cdot \bar{v}_1} \right) \bar{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} \quad \bar{v}_2' = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \bar{v}_3 &= \bar{x}_3 - \left( \frac{\bar{x}_3 \cdot \bar{v}_1}{\bar{v}_1 \cdot \bar{v}_1} \right) \bar{v}_1 - \left( \frac{\bar{x}_3 \cdot \bar{v}_2'}{\bar{v}_2' \cdot \bar{v}_2'} \right) \bar{v}_2' \\ &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{12} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} - \begin{bmatrix} -1/2 \\ 1/6 \\ 1/6 \\ 1/6 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix} \quad \bar{v}_3' = \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$Q = \begin{bmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{bmatrix}$$

4. (a) (5 points) How is the  $QR$  factorization of a matrix  $A$  used to determine the eigenvalues of  $A$ ?

Let  $A = A^{(0)} = Q_0 R_0$ .

Compute  $A^{(1)} = R_0 Q_0$ . Find  $A^{(1)} = Q_1 R_1$ .

In general, compute  $A^{(k+1)} = R_k Q_k$ . Find  $A^{(k+1)} = Q_{k+1} R_{k+1}$ .

- (b) (4 points) Without providing proof, what does the iteration described in (a) converge to?

$A^{(k+1)}$  converges to an upper triangular matrix whose eigenvalues are thus on the diagonal. Since the matrix  $A^{(k+1)}$  is similar to  $A$ , these eigenvalues are also the eigenvalues of  $A$ .

5. (a) (2 points) Let  $\mathbf{u}$  be a nonzero vector in  $\mathbb{R}^n$ . Define a Householder matrix  $H$ .

$$H = I - \frac{2\bar{\mathbf{u}}\bar{\mathbf{u}}^T}{\bar{\mathbf{u}}^T\bar{\mathbf{u}}}$$

- (b) (10 points) Let  $A = \begin{bmatrix} 2 & -1 & -2 \\ -4 & 6 & 3 \\ -4 & -2 & 8 \end{bmatrix}$ . Complete the following steps that use Householder matrices to find the matrix  $R$  in the  $QR$  factorization of  $A$  (you do not have to find the matrix  $Q$  or  $R$  - just complete the requested steps).

- Multiply  $A$  by a Householder matrix  $H_1$  to obtain  $A^{(1)} = \begin{bmatrix} -6 & 3 & 8 \\ 0 & 4 & -2 \\ 0 & -4 & 3 \end{bmatrix}$ .

$$\bar{\mathbf{x}} = \begin{bmatrix} 2 \\ -4 \\ -4 \end{bmatrix}$$

$$\bar{\mathbf{u}} = \bar{\mathbf{x}} + \sqrt{36} \bar{\mathbf{e}}_1 = \begin{bmatrix} 2 \\ -4 \\ -4 \end{bmatrix} + \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ -4 \end{bmatrix}$$

$$\begin{aligned} A^{(1)} &= H_1 A = A - \frac{2}{\bar{\mathbf{u}}^T \bar{\mathbf{u}}} \bar{\mathbf{u}} (A^T \bar{\mathbf{u}})^T \\ &= \begin{bmatrix} 2 & -1 & -2 \\ -4 & 6 & 3 \\ -4 & -2 & 8 \end{bmatrix} - \frac{2}{96} \begin{bmatrix} 8 \\ -4 \\ -4 \end{bmatrix} \left( \begin{bmatrix} 2 & -4 & -4 \\ -1 & 6 & -2 \\ -2 & 3 & 8 \end{bmatrix} \begin{bmatrix} 8 \\ -4 \\ -4 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 2 & -1 & -2 \\ -4 & 6 & 3 \\ -4 & -2 & 8 \end{bmatrix} - \frac{1}{48} \begin{bmatrix} 8 \\ -4 \\ -4 \end{bmatrix} \begin{bmatrix} 48 & -24 & -60 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 & -2 \\ -4 & 6 & 3 \\ -4 & -2 & 8 \end{bmatrix} - \begin{bmatrix} 8 \\ -4 \\ -4 \end{bmatrix} \begin{bmatrix} 1 & -1/2 & -5/4 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 & -2 \\ -4 & 6 & 3 \\ -4 & -2 & 8 \end{bmatrix} - \begin{bmatrix} 8 & -4 & -10 \\ -4 & 2 & 5 \\ -4 & 2 & 5 \end{bmatrix} = \begin{bmatrix} -6 & 3 & 8 \\ 0 & 4 & -2 \\ 0 & -4 & 3 \end{bmatrix} \end{aligned}$$

- What is the vector  $\mathbf{u}_2$  that would be used to zero out the entry in row 3, column 2 of  $A^{(1)}$ ?

$$\bar{\mathbf{x}}_2 = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\bar{\mathbf{u}}_2 = \begin{bmatrix} 4 \\ -4 \end{bmatrix} + \sqrt{32} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 + \sqrt{32} \\ -4 \end{bmatrix}$$

- Without explicitly forming the required matrices, how would the matrix  $Q$  be determined? You do not have to compute  $Q$ .

$$Q = H_1 H_2$$

6. (a) (3 points) Define what it means for an  $n \times n$  matrix  $A$  to be orthogonally diagonalizable.

$A$  is orthogonally diagonalizable if there exist an orthogonal matrix  $Q$  and diagonal matrix  $D$  such that  $Q^T A Q = D$

- (b) (8 points) Suppose  $A$  is symmetric. Let  $\lambda_1 \neq \lambda_2$  be two eigenvalues of  $A$  and suppose that  $\mathbf{v}_1$  is in the eigenspace for  $\lambda_1$  and  $\mathbf{v}_2$  is in the eigenspace for  $\lambda_2$ . Prove that  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are orthogonal.

Note that

$$\begin{aligned}\lambda_1 (\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2) &= (\lambda_1 \bar{\mathbf{v}}_1) \cdot \bar{\mathbf{v}}_2 \\ &= (A \bar{\mathbf{v}}_1) \cdot \bar{\mathbf{v}}_2 \\ &= (A \bar{\mathbf{v}}_1)^T \bar{\mathbf{v}}_2 \\ &= (\bar{\mathbf{v}}_1^T A^T) \bar{\mathbf{v}}_2 \\ &= \bar{\mathbf{v}}_1^T A \bar{\mathbf{v}}_2 \\ &= \bar{\mathbf{v}}_1^T (\lambda_2 \bar{\mathbf{v}}_2) \\ &= \lambda_2 (\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2)\end{aligned}$$

Therefore  $\lambda_1 (\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2) - \lambda_2 (\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2) = 0$  or equivalently  $(\lambda_1 - \lambda_2) \bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2 = 0$ .

Since  $\lambda_1 - \lambda_2 \neq 0$ ,  $\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2 = 0$ . Thus,  $\bar{\mathbf{v}}_1$  and  $\bar{\mathbf{v}}_2$  are orthogonal.

7. (12 points) Let  $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ . The characteristic equation of  $A$  is  $c_A(\lambda) = -(\lambda - 7)^2(\lambda + 2)$ . The eigenspace for  $\lambda = -2$  and is given by

$$E_{-2} = \text{Span} \left\{ \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

Find the orthogonal diagonalization of  $A$ . Clearly state the matrices  $Q$  and  $D$ .

$$\underline{\lambda = 7}$$

$$\begin{bmatrix} -4 & -2 & 4 \\ -2 & -1 & 2 \\ 4 & 2 & -4 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$2x + y - 2z = 0$$

$$x = -\frac{1}{2}s + t$$

$$2x = -y + 2z$$

$$y = s$$

$$x = -\frac{1}{2}s + z$$

$$z = t$$

$$\vec{x} = \begin{bmatrix} -\frac{1}{2}s + t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$E_7 = \text{Span} \left\{ \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Now use Gram-Schmidt to find orthogonal basis

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} - \left( \frac{\begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}} \right) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 2 \\ 1/2 \end{bmatrix}$$

$$E_7 = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} \right\}$$

$$Q = \begin{bmatrix} -2/3 & 1/\sqrt{2} & -1/\sqrt{18} \\ -1/3 & 0 & 4/\sqrt{18} \\ 2/3 & 1/\sqrt{2} & 1/\sqrt{18} \end{bmatrix}$$

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

8. (a) (4 points) Identify the matrix  $A$  associated with the quadratic form  $Q(\mathbf{x}) = x^2 + 7y^2 - 3z^2 + xy - 2xz + 3yz$ .

$$A = \begin{bmatrix} 1 & 1/2 & -1 \\ 1/2 & 7 & 3/2 \\ -1 & 3/2 & -3 \end{bmatrix}$$

- (b) (6 points) Consider the quadratic form  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$  where  $A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$ . Classify  $Q(\mathbf{x})$  as one of positive semidefinite, negative semidefinite or indefinite.

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -4 \\ -4 & -5-\lambda \end{vmatrix} = (1-\lambda)(-5-\lambda) - 16 = \lambda^2 + 5\lambda - \lambda - 5 - 16 = \lambda^2 + 4\lambda - 21 = (\lambda+7)(\lambda-3)$$

Thus, the eigenvalues of  $A$  are  $-7$  and  $3$  and  $A$  is indefinite

9. (8 points) Let  $A$  be an  $n \times n$  symmetric matrix. Prove that there exists an orthogonal change of variables  $\mathbf{x} = P\mathbf{y}$  that transforms the quadratic form  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$  into a quadratic form with no cross-product terms.

Since  $A$  is symmetric  $A$  is orthogonally diagonalizable. Thus there exists an orthogonal matrix  $P$  and diagonal matrix  $D$  such that  $A = P D P^T$ . Let  $\bar{\mathbf{x}} = P\bar{\mathbf{y}}$ .

$$\text{Then, } \bar{\mathbf{y}} = P^{-1}\bar{\mathbf{x}} = P^T \bar{\mathbf{x}}$$

$$\begin{aligned} Q(\bar{\mathbf{x}}) &= \bar{\mathbf{x}}^T A \bar{\mathbf{x}} \\ &= \bar{\mathbf{x}}^T P D P^T \bar{\mathbf{x}} \\ &= (P^T \bar{\mathbf{x}})^T D (P^T \bar{\mathbf{x}}) \\ &= \bar{\mathbf{y}}^T D \bar{\mathbf{y}} \end{aligned}$$

which is a quadratic form with no cross product terms.

10. (12 points) Let  $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ . The eigenvalues of  $A$  are  $\lambda = 0, 4, 9$  with corresponding eigenvectors

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

respectively. Find the singular value decomposition of  $A$ . Be sure to clearly state  $U$ ,  $V$ , and  $\Sigma$ .

$$\bar{V}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \bar{V}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \bar{V}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\bar{u}_1 = \frac{1}{\sigma_1} A \bar{V}_1 = \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$\bar{u}_2 = \frac{1}{\sigma_2} A \bar{V}_2 = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Need a vector  $\bar{u}_3$  with  $\bar{u}_1 \cdot \bar{u}_3 = 0$  and  $\bar{u}_2 \cdot \bar{u}_3 = 0$ . If  $\bar{u}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$  we have

$$\begin{aligned} -y &= 0 \\ x + z &= 0 \end{aligned}$$

Thus,

$$\bar{u}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Therefore,

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix} \\ &= U \Sigma V^T \end{aligned}$$

where

$$U = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \quad \Sigma = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad V = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$